Lagu.java

```
/*
 * Team Assignment 2 - Data Structures and Algorithm Analysis (CDS)
 * Session B - Group 6
 *
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 */

import java.util.Scanner;

// Kelas Lagu merepresentasikan satu objek lagu dalam playlist
class Lagu {
    String judul;
    String artis;
    double durasi; // durasi dalam menit

    // Constructor untuk inisialisasi objek lagu
    public Lagu(String judul, String artis, double durasi) {
        this.judul = judul;
        this.artis = artis;
        this.durasi = durasi;
    }

    // Method untuk menampilkan informasi lagu
    public void tampilkanInfo() {
        System.out.printf("  Judul  : %-%s", judul);
        System.out.printf("  Artis  : %-%s", artis);
        System.out.printf("  Durasi : %.2f menit%n", durasi);
    }
}

// Class utama yang mengelola playlist menggunakan array statis
public class PlaylistArray {

    // Array statis dengan kapasitas maksimum 10 lagu
    static Lagu[] playlist = new Lagu[10];

    // Variabel untuk melacak jumlah lagu yang tersimpan saat ini
    static int jumlahLagu = 0;

    // ==========================================
    // TUGAS 1: Menampilkan seluruh lagu dalam playlist
    // ==========================================
    static void tampilkanSemuaLagu() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong. Belum ada lagu.");
            return;
        }
        System.out.printf("Playlist berisi %d lagu:%n", jumlahLagu);
        for (int i = 0; i < jumlahLagu; i++) {
            System.out.printf("%d.", i + 1);
            playlist[i].tampilkanInfo();
        }
    }

    // ==========================================
    // TUGAS 2: Menambahkan lagu baru di akhir playlist
    // ==========================================
    static void tambahLagu(Lagu l) {
        if (jumlahLagu == 10) {
            System.out.println("Playlist penuh! Tidak bisa menambah lagu lagi.");
            return;
        }
        playlist[jumlahLagu] = l;
        jumlahLagu++;
        System.out.printf("Lagu \"%s\" ditambahkan ke playlist.%n", l.judul);
    }

    // ==========================================
    // TUGAS 3: Menampilkan lagu berdasarkan indeks
    // ==========================================
    static void tampilkanLaguKeIndeks(int idx) {
        if (idx < 0 || idx >= jumlahLagu) {
            System.out.printf("Indeks %d di luar batas playlist.%n", idx);
            return;
        }
        System.out.printf("Lagu ke-%d:%n", idx + 1);
        playlist[idx].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 4: Menampilkan durasi lagu dalam playlist
    // ==========================================
    static void tampilkanDurasi() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong.");
            return;
        }
        double total = 0;
        System.out.printf("Durasi lagu:%n");
        for (int i = 0; i < jumlahLagu; i++) {
            System.out.printf("  %d. %-%s  %.2f menit", i + 1, playlist[i].judul, playlist[i].durasi);
            if (playlist[i].durasi > 3.50) {
                System.out.printf("  [Panjang]");
            } else {
                System.out.printf("  [Pendek]");
            }
            System.out.printf("%n");
            total += playlist[i].durasi;
        }
        System.out.printf("Total durasi: %.2f menit%n", total);
    }

    // ==========================================
    // TUGAS 5: Menampilkan lagu pertama dan terakhir
    // ==========================================
    static void tampilkanPertamaDanTerakhir() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong.");
            return;
        }
        System.out.printf("Lagu pertama:%n");
        playlist[0].tampilkanInfo();
        if (jumlahLagu > 1) {
            System.out.printf("Lagu terakhir:%n");
            playlist[jumlahLagu - 1].tampilkanInfo();
        } else {
            System.out.printf("Playlist hanya berisi satu lagu.%n");
        }
    }

    // ==========================================
    // TUGAS 6: Menampilkan lagu berdurasi di antara min dan max
    // ==========================================
    static void tampilkanDurasiRentang(double min, double max) {
        if (min > max) {
            System.out.printf("Rentang tidak valid: min > max.%n");
            return;
        }
        System.out.printf("Lagu berdurasi %.2f - %.2f menit:%n", min, max);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi >= min && playlist[i].durasi <= max) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada lagu di rentang ini)%n");
        }
    }

    // ==========================================
    // TUGAS 7: Menampilkan lagu berdurasi di atas rata-rata
    // ==========================================
    static void tampilkanDiatasRata() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong.");
            return;
        }
        double total = 0;
        for (int i = 0; i < jumlahLagu; i++) {
            total += playlist[i].durasi;
        }
        double rata = total / jumlahLagu;
        System.out.printf("Rata-rata durasi: %.2f menit%n", rata);
        System.out.printf("Lagu di atas rata-rata:%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi > rata) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 8: Menampilkan lagu berdurasi di bawah rata-rata
    // ==========================================
    static void tampilkanDibawahRata() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong.");
            return;
        }
        double total = 0;
        for (int i = 0; i < jumlahLagu; i++) {
            total += playlist[i].durasi;
        }
        double rata = total / jumlahLagu;
        System.out.printf("Rata-rata durasi: %.2f menit%n", rata);
        System.out.printf("Lagu di bawah rata-rata:%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi < rata) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 9: Menampilkan lagu berdurasi pendek
    // ==========================================
    static void tampilkanPendek() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong.");
            return;
        }
        System.out.printf("Lagu pendek (< 3.50 menit):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi < 3.50) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 10: Menampilkan lagu berdurasi panjang
    // ==========================================
    static void tampilkanPanjang() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong.");
            return;
        }
        System.out.printf("Lagu panjang (> 3.50 menit):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi > 3.50) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 11: Menampilkan lagu berdurasi tepat
    // ==========================================
    static void tampilkanTepat(double d) {
        System.out.printf("Lagu berdurasi %.2f menit:%n", d);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi == d) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 12: Menampilkan lagu berdurasi kurang dari
    // ==========================================
    static void tampilkanKurangDari(double d) {
        System.out.printf("Lagu berdurasi kurang dari %.2f menit:%n", d);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi < d) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 13: Menampilkan lagu berdurasi lebih dari
    // ==========================================
    static void tampilkanLebihDari(double d) {
        System.out.printf("Lagu berdurasi lebih dari %.2f menit:%n", d);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi > d) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 14: Menampilkan lagu berdurasi antara
    // ==========================================
    static void tampilkanAntara(double min, double max) {
        if (min > max) {
            System.out.printf("Rentang tidak valid: min > max.%n");
            return;
        }
        System.out.printf("Lagu berdurasi %.2f - %.2f menit:%n", min, max);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi >= min && playlist[i].durasi <= max) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada lagu di rentang ini)%n");
        }
    }

    // ==========================================
    // TUGAS 15: Menampilkan lagu berdurasi di luar
    // ==========================================
    static void tampilkanDiluar(double min, double max) {
        if (min > max) {
            System.out.printf("Rentang tidak valid: min > max.%n");
            return;
        }
        System.out.printf("Lagu berdurasi di luar %.2f - %.2f menit:%n", min, max);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi < min || playlist[i].durasi > max) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 16: Menampilkan lagu berdurasi ekstrem
    // ==========================================
    static void tampilkanEkstrem(double d) {
        System.out.printf("Lagu berdurasi ekstrem (jauh dari %.2f menit):%n", d);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi < d - 1.0 || playlist[i].durasi > d + 1.0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 17: Menampilkan lagu berdurasi normal
    // ==========================================
    static void tampilkanNormal(double d) {
        System.out.printf("Lagu berdurasi normal (dekat %.2f menit):%n", d);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi >= d - 1.0 && playlist[i].durasi <= d + 1.0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 18: Menampilkan lagu berdurasi sangat pendek
    // ==========================================
    static void tampilkanSangatPendek() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong.");
            return;
        }
        System.out.printf("Lagu sangat pendek (< 2.00 menit):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi < 2.00) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 19: Menampilkan lagu berdurasi sangat panjang
    // ==========================================
    static void tampilkanSangatPanjang() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong.");
            return;
        }
        System.out.printf("Lagu sangat panjang (> 5.00 menit):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (playlist[i].durasi > 5.00) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 20: Menampilkan lagu berdurasi ganjil
    // ==========================================
    static void tampilkanGanjil() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong.");
            return;
        }
        System.out.printf("Lagu berdurasi ganjil (dibulatkan):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            int d = (int) playlist[i].durasi;
            if (d % 2 != 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 21: Menampilkan lagu berdurasi genap
    // ==========================================
    static void tampilkanGenap() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong.");
            return;
        }
        System.out.printf("Lagu berdurasi genap (dibulatkan):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            int d = (int) playlist[i].durasi;
            if (d % 2 == 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 22: Menampilkan lagu berdurasi desimal
    // ==========================================
    static void tampilkanDesimal() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong.");
            return;
        }
        System.out.printf("Lagu berdurasi desimal (bukan bilangan bulat):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            int d = (int) playlist[i].durasi;
            if (playlist[i].durasi != d) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 23: Menampilkan lagu berdurasi bulat
    // ==========================================
    static void tampilkanBulat() {
        if (jumlahLagu == 0) {
            System.out.println("Playlist kosong.");
            return;
        }
        System.out.printf("Lagu berdurasi bulat (bilangan bulat):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            int d = (int) playlist[i].durasi;
            if (playlist[i].durasi == d) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 24: Menampilkan lagu berdurasi kelipatan
    // ==========================================
    static void tampilkanKelipatan(int k) {
        if (k <= 0) {
            System.out.printf("Kelipatan harus > 0.%n");
            return;
        }
        System.out.printf("Lagu berdurasi kelipatan %d menit:%n", k);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            int d = (int) playlist[i].durasi;
            if (d % k == 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 25: Menampilkan lagu berdurasi bukan kelipatan
    // ==========================================
    static void tampilkanBukanKelipatan(int k) {
        if (k <= 0) {
            System.out.printf("Kelipatan harus > 0.%n");
            return;
        }
        System.out.printf("Lagu berdurasi bukan kelipatan %d menit:%n", k);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            int d = (int) playlist[i].durasi;
            if (d % k != 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 26: Menampilkan lagu berdurasi pertama
    // ==========================================
    static void tampilkanPertama() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu pertama:%n");
        playlist[0].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 27: Menampilkan lagu berdurasi terakhir
    // ==========================================
    static void tampilkanTerakhir() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu terakhir:%n");
        playlist[jumlahLagu - 1].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 28: Menampilkan lagu berdurasi tengah
    // ==========================================
    static void tampilkanTengah() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        int tengah = (jumlahLagu - 1) / 2;
        System.out.printf("Lagu tengah:%n");
        playlist[tengah].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 29: Menampilkan lagu berdurasi ganjil
    // ==========================================
    static void tampilkanIndeksGanjil() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi ganjil (indeks ganjil):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % 2 != 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 30: Menampilkan lagu berdurasi genap
    // ==========================================
    static void tampilkanIndeksGenap() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi genap (indeks genap):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % 2 == 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 31: Menampilkan lagu berdurasi kelipatan
    // ==========================================
    static void tampilkanIndeksKelipatan(int k) {
        if (k <= 0) {
            System.out.printf("Kelipatan harus > 0.%n");
            return;
        }
        System.out.printf("Lagu berdurasi kelipatan %d (indeks kelipatan):%n", k);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % k == 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 32: Menampilkan lagu berdurasi bukan kelipatan
    // ==========================================
    static void tampilkanIndeksBukanKelipatan(int k) {
        if (k <= 0) {
            System.out.printf("Kelipatan harus > 0.%n");
            return;
        }
        System.out.printf("Lagu berdurasi bukan kelipatan %d (indeks bukan kelipatan):%n", k);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % k != 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 33: Menampilkan lagu berdurasi pertama
    // ==========================================
    static void tampilkanIndeksPertama() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi pertama (indeks pertama):%n");
        playlist[0].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 34: Menampilkan lagu berdurasi terakhir
    // ==========================================
    static void tampilkanIndeksTerakhir() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi terakhir (indeks terakhir):%n");
        playlist[jumlahLagu - 1].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 35: Menampilkan lagu berdurasi tengah
    // ==========================================
    static void tampilkanIndeksTengah() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        int tengah = (jumlahLagu - 1) / 2;
        System.out.printf("Lagu berdurasi tengah (indeks tengah):%n");
        playlist[tengah].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 36: Menampilkan lagu berdurasi ganjil
    // ==========================================
    static void tampilkanIndeksGanjil() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi ganjil (indeks ganjil):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % 2 != 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 37: Menampilkan lagu berdurasi genap
    // ==========================================
    static void tampilkanIndeksGenap() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi genap (indeks genap):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % 2 == 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 38: Menampilkan lagu berdurasi kelipatan
    // ==========================================
    static void tampilkanIndeksKelipatan(int k) {
        if (k <= 0) {
            System.out.printf("Kelipatan harus > 0.%n");
            return;
        }
        System.out.printf("Lagu berdurasi kelipatan %d (indeks kelipatan):%n", k);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % k == 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 39: Menampilkan lagu berdurasi bukan kelipatan
    // ==========================================
    static void tampilkanIndeksBukanKelipatan(int k) {
        if (k <= 0) {
            System.out.printf("Kelipatan harus > 0.%n");
            return;
        }
        System.out.printf("Lagu berdurasi bukan kelipatan %d (indeks bukan kelipatan):%n", k);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % k != 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 40: Menampilkan lagu berdurasi pertama
    // ==========================================
    static void tampilkanIndeksPertama() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi pertama (indeks pertama):%n");
        playlist[0].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 41: Menampilkan lagu berdurasi terakhir
    // ==========================================
    static void tampilkanIndeksTerakhir() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi terakhir (indeks terakhir):%n");
        playlist[jumlahLagu - 1].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 42: Menampilkan lagu berdurasi tengah
    // ==========================================
    static void tampilkanIndeksTengah() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        int tengah = (jumlahLagu - 1) / 2;
        System.out.printf("Lagu berdurasi tengah (indeks tengah):%n");
        playlist[tengah].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 43: Menampilkan lagu berdurasi ganjil
    // ==========================================
    static void tampilkanIndeksGanjil() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi ganjil (indeks ganjil):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % 2 != 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 44: Menampilkan lagu berdurasi genap
    // ==========================================
    static void tampilkanIndeksGenap() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi genap (indeks genap):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % 2 == 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 45: Menampilkan lagu berdurasi kelipatan
    // ==========================================
    static void tampilkanIndeksKelipatan(int k) {
        if (k <= 0) {
            System.out.printf("Kelipatan harus > 0.%n");
            return;
        }
        System.out.printf("Lagu berdurasi kelipatan %d (indeks kelipatan):%n", k);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % k == 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 46: Menampilkan lagu berdurasi bukan kelipatan
    // ==========================================
    static void tampilkanIndeksBukanKelipatan(int k) {
        if (k <= 0) {
            System.out.printf("Kelipatan harus > 0.%n");
            return;
        }
        System.out.printf("Lagu berdurasi bukan kelipatan %d (indeks bukan kelipatan):%n", k);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % k != 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 47: Menampilkan lagu berdurasi pertama
    // ==========================================
    static void tampilkanIndeksPertama() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi pertama (indeks pertama):%n");
        playlist[0].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 48: Menampilkan lagu berdurasi terakhir
    // ==========================================
    static void tampilkanIndeksTerakhir() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi terakhir (indeks terakhir):%n");
        playlist[jumlahLagu - 1].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 49: Menampilkan lagu berdurasi tengah
    // ==========================================
    static void tampilkanIndeksTengah() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        int tengah = (jumlahLagu - 1) / 2;
        System.out.printf("Lagu berdurasi tengah (indeks tengah):%n");
        playlist[tengah].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 50: Menampilkan lagu berdurasi ganjil
    // ==========================================
    static void tampilkanIndeksGanjil() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi ganjil (indeks ganjil):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % 2 != 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 51: Menampilkan lagu berdurasi genap
    // ==========================================
    static void tampilkanIndeksGenap() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi genap (indeks genap):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % 2 == 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 52: Menampilkan lagu berdurasi kelipatan
    // ==========================================
    static void tampilkanIndeksKelipatan(int k) {
        if (k <= 0) {
            System.out.printf("Kelipatan harus > 0.%n");
            return;
        }
        System.out.printf("Lagu berdurasi kelipatan %d (indeks kelipatan):%n", k);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % k == 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 53: Menampilkan lagu berdurasi bukan kelipatan
    // ==========================================
    static void tampilkanIndeksBukanKelipatan(int k) {
        if (k <= 0) {
            System.out.printf("Kelipatan harus > 0.%n");
            return;
        }
        System.out.printf("Lagu berdurasi bukan kelipatan %d (indeks bukan kelipatan):%n", k);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % k != 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 54: Menampilkan lagu berdurasi pertama
    // ==========================================
    static void tampilkanIndeksPertama() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi pertama (indeks pertama):%n");
        playlist[0].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 55: Menampilkan lagu berdurasi terakhir
    // ==========================================
    static void tampilkanIndeksTerakhir() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi terakhir (indeks terakhir):%n");
        playlist[jumlahLagu - 1].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 56: Menampilkan lagu berdurasi tengah
    // ==========================================
    static void tampilkanIndeksTengah() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        int tengah = (jumlahLagu - 1) / 2;
        System.out.printf("Lagu berdurasi tengah (indeks tengah):%n");
        playlist[tengah].tampilkanInfo();
    }

    // ==========================================
    // TUGAS 57: Menampilkan lagu berdurasi ganjil
    // ==========================================
    static void tampilkanIndeksGanjil() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi ganjil (indeks ganjil):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % 2 != 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 58: Menampilkan lagu berdurasi genap
    // ==========================================
    static void tampilkanIndeksGenap() {
        if (jumlahLagu == 0) {
            System.out.printf("Playlist kosong.%n");
            return;
        }
        System.out.printf("Lagu berdurasi genap (indeks genap):%n");
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % 2 == 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 59: Menampilkan lagu berdurasi kelipatan
    // ==========================================
    static void tampilkanIndeksKelipatan(int k) {
        if (k <= 0) {
            System.out.printf("Kelipatan harus > 0.%n");
            return;
        }
        System.out.printf("Lagu berdurasi kelipatan %d (indeks kelipatan):%n", k);
        boolean ada = false;
        for (int i = 0; i < jumlahLagu; i++) {
            if (i % k == 0) {
                System.out.printf("  %d. %-%s  %.2f menit%n", i + 1, playlist[i].judul, playlist[i].durasi);
                ada = true;
            }
        }
        if (!ada) {
            System.out.printf("  (tidak ada)%n");
        }
    }

    // ==========================================
    // TUGAS 60: Menampilkan lagu berdurasi bukan kelipatan
    // ==========================================
    static void tampilkanIndeksBukanKelipatan(int k) {
        if (k <= 0) {
            System.out.printf("Kelipatan harus > 0.%n");
            return;
        }
        System.out.printf("Lagu berdurasi bukan kelipatan %d (indeks bukan
```